Week 51 Honest measurement ranges

Facilitator guide | Final student packets v3 | Prepared 3 October 2026 | Revised 23 October 2026 | Unpiloted

Mathematical overview

Guaranteed intervals. If $a \in [L,U]$ and $b \in [V,W]$, then $a+b \in [L+V,U+W]$ and $a-b \in [L-W,U-V]$. When a and b may vary freely in their separate closed intervals, these bounds are tight and both endpoints can occur. This is deterministic uncertainty: no probability distribution or confidence percentage is assumed.

“Always reaches r” means the minimum is at least r; “always fits within s” means the maximum is at most s. Equality is allowed in both. Except when a problem explicitly asks for whole-unit pairs, hidden endpoints may lie anywhere in the band, including between ticks and at the edges. The cards stipulate guaranteed bounds; the ruler itself does not establish an instrument error bound.

Dependencies matter. Repeated A is one unchanged quantity. Thus $(A+4)-(A+1)=3$ and $(A+B)-A=B$. Using separate ranges for repeated copies can enclose the answer too broadly. A known relation such as $A+B=12$ restricts possible endpoint combinations. The elementary formulas remain enclosures, but need not remain tight when the variables are linked.

Discovery by band. K-1 builds extreme examples and tries to defeat “always” claims. Grades 2-3 designs the widest safe card and compares shared versus separate strips. Grades 4-5 tightens bounds using an exact total and shared-piece cancellation. Testing examples can disprove a claim; endpoint arguments prove a guarantee and its sharpness.

Prepare and launch

Per pair or trio: printed pages, pencils and one strip set from **week-51-materials.pdf** on card stock at actual size (two sheets for a table of two pairs). **One unit is 10 mm**; each main 0-16 ruler and each sheet row is 160 mm long. A set has two gray strips of each whole length from 1 to 15 units and two white add-ons each of 4 and 1 units for the shared-A page; every extreme in the keys is a whole length. To take a length from a band, lay a strip along it from 0 and check that the end lands inside; join strips end to end on a 0-16 ruler. Repeated A means two equal strips, or one strip used twice. Cuisenaire rods also fit. Cutting takes about 10 minutes per set, less with a paper trimmer; the sets are reusable. The kit is physically unrehearsed.

Readiness. K-1 needs matching endpoints and counting small lengths, with adult reading. Older children need addition/subtraction and comparison of intervals. Negative numbers are unnecessary: in every printed gap task, the designated upper/longer join stays to the right. Keep an adult at each of the three tables.

Launch, 4 minutes. Let children find the strips whose ends land inside a band. Show A from 2 to 3 and B from 1 to 2 using the printed visual: A=2 and B=2, joined gap-free from 0, end at 4, inside the A + B band. Ask for a shorter join and a longer one; the shortest is 3 and the longest 5. Ask, “Can you choose allowed lengths that break my promise?”

Flexible hour. 0-5 arrival; 5-9 launch; 9-47 tables: P1-2 with partner tests, then P3 (older bands also P4) or return pages; 47-53 each pair shows one guarantee and the two settings at its ends; 53-60 reset. Keep uncertain ranges visibly distinct from one actual trial.

Key for the join and card problems

Problem 1, all bands. A is 4-6 and B is 3-4, so the tight join range is **[7,10]**. Extremes are 4+3=7 and 6+4=10. K-1 can make exactly 8 with 4+4 or 5+3. In the older task, either extreme defeats the claimed 8-9 guarantee. Every sum lies in [7,10], and both endpoints are allowed, so neither bound can improve.

Problem 2. Use only the three printed row pairs, not arbitrary cross-pairings.

Printed pair	Tight sum	Always reaches 8	Also fits within 10
A 4-6 with B 3-4	[7,10]	No: 4+3=7	Yes
C 4-5 with D 4-5	[8,10]	Yes	Yes
E 6-8 with F 2-4	[8,12]	Yes	No: 8+4=12

K-1 asks only about reaching 8, so **C with D and E with F** both work. Grades 2-3 and 4-5 require both promises, so **C with D alone** works. Reaching exactly 8 and ending exactly 10 count as successes. The endpoint witnesses in the table defeat the rejected guarantees.

Problem 3. Choose **B from 4 to 5**. With A from 3 to 4, all sums then lie in [7,9]. K-1's 3-4 card fails at 3+3=6; its 5-6 card fails at 4+6=10. In the older task, B's lower endpoint must be at least 4, because A might be 3; B's upper endpoint must be at most 5, because A might be 4. Hence [4,5] is the unique widest interval.

Grades 2-3 and 4-5 Problem 4. Uncover **A**. Knowing A=a leaves [a+3,a+4], width 1. Knowing B=b leaves [b+4,b+6], width 2. Revealing the strip with the wider original range removes more uncertainty; the exact revealed value shifts the remaining range but does not change its width.

Key for gaps between two strips

K-1 Problem 4. The long strip is 7-9 and the short strip 5-6. Largest gap **4** occurs at 9-5; smallest gap **1** occurs at 7-6. Each end moves separately. Subtracting shortest from shortest would not give the smallest gap.

K-1 Problem 5. Two whole-unit ways to make gap 3 are **8-5 and 9-6**. With continuous endpoints, there are also intermediate choices such as 8.5-5.5; the prompt does not restrict this problem to whole units.

Grades 2-3 and 4-5 Problem 5. Long strip [13,15] minus short strip [10,12] has tight gap **[1,5]**. Use 13-12=1 and 15-10=5 to show both bounds are attained. Every allowed long length is at least 13 and every short one at most 12, so the difference is positive.

Grades 2-3 and 4-5 Problem 6. Subtracting middle values gives 14-11=3, which is possible but not guaranteed. Either 13 and 12 (gap 1) or 15 and 10 (gap 5) defeats the assertion of one exact gap. No probabilistic claim makes the midpoint more authoritative.

Key for shared and separate strips

K-1 Problem 6; older Problem 7. Top comparison: A+4 versus that same A+1. The end gap is always **3**; both smallest and largest gaps are 3, because the shared A cancels.

Bottom comparison: A+4 versus B+1, with A and B each separately allowed from 4 to 6. The tight gap is **[1,5]**. Minimum: A=4,B=6 gives 8-7=1. Maximum: A=6,B=4 gives 10-5=5. The two extremes use different values of A, since each try may use new lengths. Intermediate values are possible too. The diagram shows one setting, not a promise that the shaded pieces always retain their drawn lengths.

Key for exact totals

K-1 Problem 7; Grades 2-3 Problem 8. With A from 3 to 7, B from 3 to 5, whole-unit lengths, and total 9, the complete list of ordered pairs (A,B) is **(4,5), (5,4), (6,3)**. A=3 would need B=6 and A=7 would need B=2, both off B's card; for A=4, 5 and 6, B=9-A fits its card. Four boards are printed for three pairs, so the child decides when the list is complete. The exact-sum condition links the strips.

Grades 4-5 Problem 8. A∈[7,10], B∈[3,7] and A+B=12 imply B=12-A. The B card restricts A to [5,9]; intersecting with A's card gives **A∈[7,9]** and consequently B∈[3,5]. Since A is always larger, the end gap is A-B=2A-12, ranging from **2 to 6**. Witnesses are (A,B)=(7,5) and (9,3). The discarded choice A=10 would require B=2 and is not allowed.

Grades 4-5 Problem 9. The tight remainder range is **[3,4]**, exactly B's original range. Physically remove the same A that was joined: all that remains is B. The claimed [1,6] is a valid but unnecessarily wide enclosure obtained by treating the total and A as freely variable. Its extreme combinations cannot occur together: a total of 7 requires A=4,B=3, while a total of 10 requires A=6,B=4.

Optional hints and source

For a guarantee, ask the partner to choose the most awkward permitted ends. For a tight bound, ask for a setting that actually reaches it. For dependency, keep a finger or matching label on the same A. If the group is still comparing physical extremes, stop there rather than require interval notation or algebra.

NIST Digital Library of Mathematical Functions, **§3.1(ii)**, "Interval Arithmetic," equation (3.1.6): official reference. This is the current interval-arithmetic section; §3.1(v) is error measures. The strip endpoint calculations, repeated-variable cancellations and constrained-total cases here are independently derived. All dimensions were digitally checked; strip handling and the hour plan are unpiloted proposals.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

First routes in all bands: P1-2, using allowed endpoint choices to test each promise; add P3 for range-card design. On return, K-1 can make gap extremes (P4-5), compare shared and separate strips (P6), or fill the exact total (P7). Grades 2-3 can find difference ranges and compare repeated pieces (P5-7), then enumerate fixed-total whole-unit pairs (P8). Grades 4-5 can tighten the linked-total range and cancel the very same piece (P8-9) after the dependency comparison in P7. Keep one unchanged A whenever a problem repeats it.